STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

15 October 2026

Self-report

To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).

Part 1

1a)

Since $\mathbf{m}$ and $\boldsymbol{\varepsilon}$ are independent and Gaussian, $(\mathbf{m}, \mathbf{d})$ is a linear transformation of a Gaussian vector and therefore jointly Gaussian, with

$$\begin{aligned} E(\mathbf{d}) &= \mathbf{A}\boldsymbol{\mu}_m, \\ \mathrm{Cov}(\mathbf{d}) &= \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I} =: \boldsymbol{\Sigma}_d, \\ \mathrm{Cov}(\mathbf{d}, \mathbf{m}) &= \mathrm{Cov}(\mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\boldsymbol{\Sigma}_m, \end{aligned}$$

so

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_m \\ \mathbf{A}\boldsymbol{\mu}_m \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_m\mathbf{A}^\top \\ \mathbf{A}\boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_d \end{pmatrix} \right).$$

The conditional distribution of a partitioned Gaussian gives the posterior $\mathbf{m} \mid \mathbf{d} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ with

$$\begin{aligned} \boldsymbol{\mu}_{m|d} &= \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m), \\ \boldsymbol{\Sigma}_{m|d} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}\mathbf{A}\boldsymbol{\Sigma}_m. \end{aligned}$$

1b)

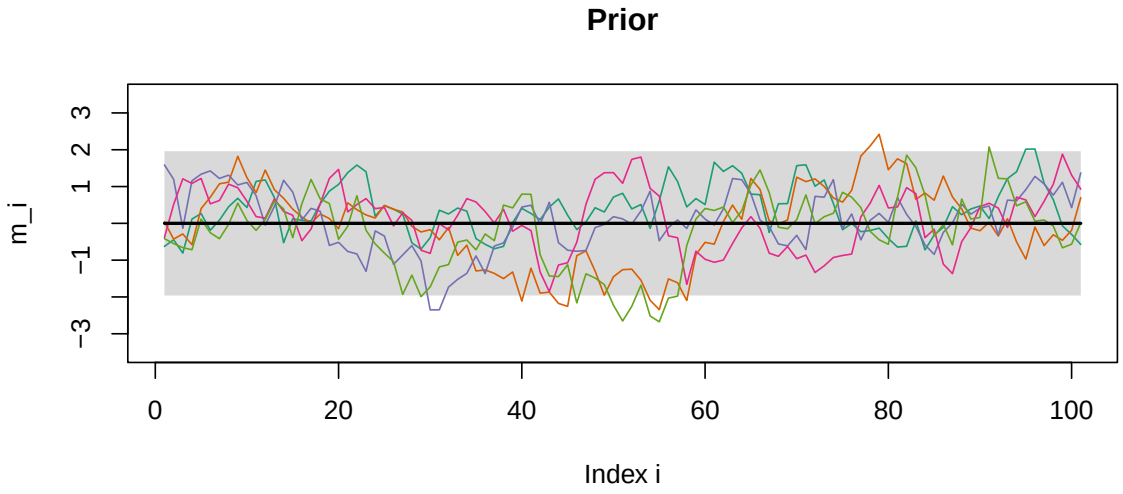


Figure 1: Prior expectation (black), 95% credibility interval (grey) and five simulated profiles.

1c)

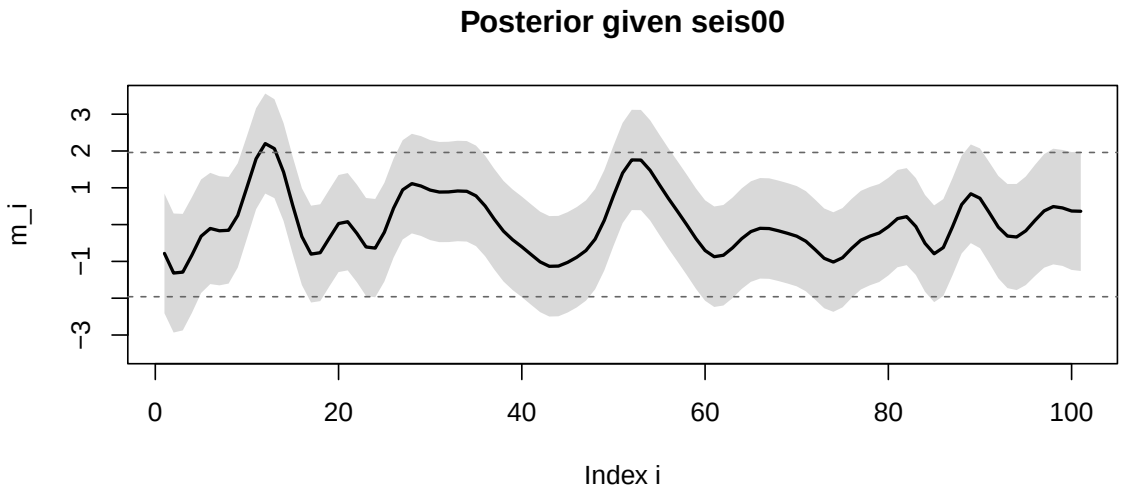


Figure 2: Posterior expectation (black) and 95% credibility interval (grey) given seis00. Dashed: prior interval.

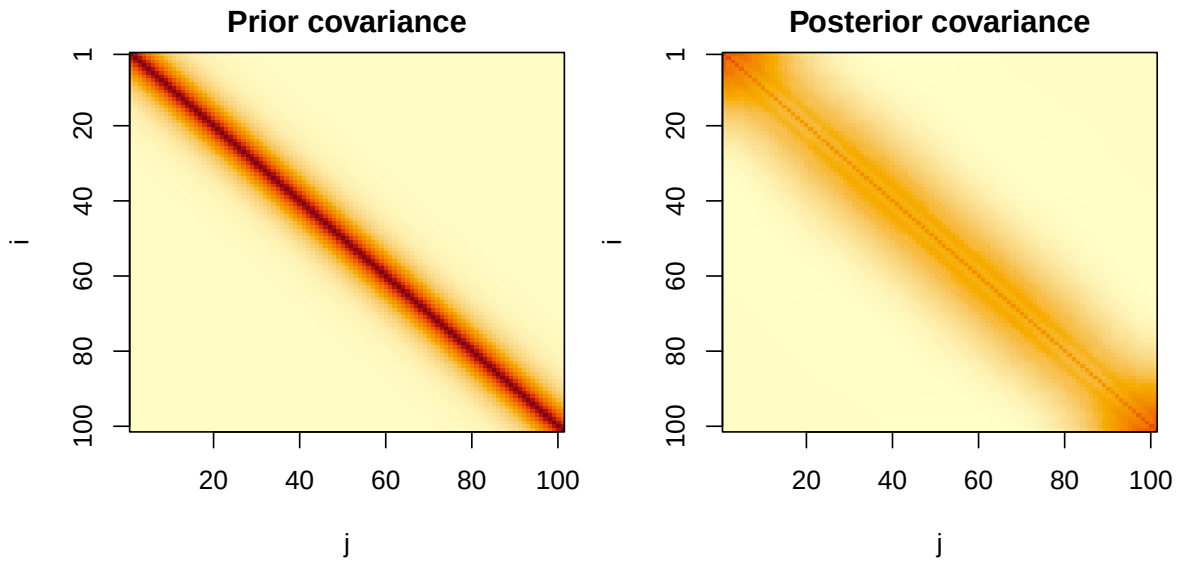


Figure 3: Prior (left) and posterior (right) covariance matrices, same colour scale.

1d)

Table 1: Posterior summaries of V ($B = 100$).

mean	median	sd
8.76	5.48	8.68

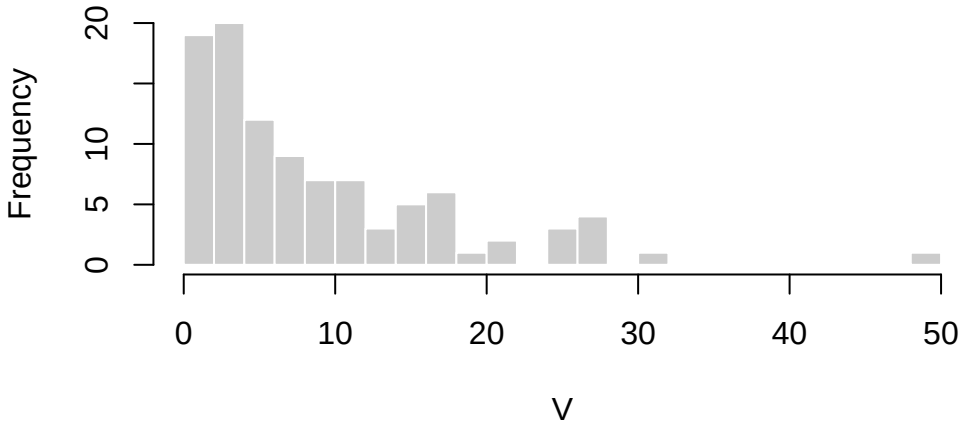


Figure 4: Histogram of 100 posterior samples of V.

1e)

Estimates from the ordered samples: $\widehat{P}_{10} = V_{(10)} = 0.92$ and $\widehat{P}_{90} = V_{(90)} = 20.64$.

The number of samples below P_{10} is $N \sim \text{Bin}(100, 0.1)$. Since $V_{(r)} \leq P_{10} < V_{(s)}$ exactly when $r \leq N \leq s-1$,

$$P\big(V_{(r)} \leq P_{10} < V_{(s)}\big) = \sum_{k=r}^{s-1} \binom{100}{k} 0.1^k 0.9^{100-k}.$$

With $r = 5$ and $s = 17$ this probability is 0.956, which gives the interval $[V_{(5)}, V_{(17)}] = [0.19,\ 1.82]$ for P_{10}. The interval is wide compared with the estimate, so $B = 100$ is not sufficient.

1f)

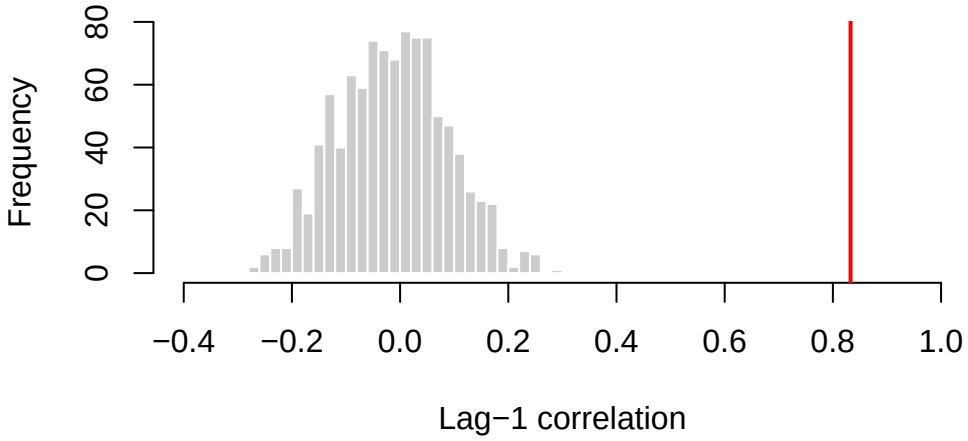


Figure 5: Lag-1 correlation for 1000 permutations of seis00 (histogram) and for the observed data (red line).

The observed lag-1 correlation $\hat{\rho}_1 = 0.83$ is larger than all 1000 permuted values ($p < 0.001$), so the claim that the data are exchangeable is rejected.

Part 2

2a)

For a given (T, B) only the prior mean changes, to $\boldsymbol{\mu}_m^{T,B}$; the covariance is as in Part 1. From 1a, the marginal distribution of the data is

$$p(\mathbf{d} \mid T, B) = N\big(\mathbf{d};\ \mathbf{A}\boldsymbol{\mu}_m^{T,B},\ \boldsymbol{\Sigma}_d\big), \qquad \boldsymbol{\Sigma}_d = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I},$$

and Bayes' rule gives, for $(T, B) \in \mathcal{G}$,

$$p(T, B \mid \mathbf{d}) = \frac{p(\mathbf{d} \mid T, B)\, p(T, B)}{\sum_{(T', B') \in \mathcal{G}} p(\mathbf{d} \mid T', B')\, p(T', B')} \ \propto\ \exp\Big\{ -\tfrac{1}{2}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B})^\top \boldsymbol{\Sigma}_d^{-1} (\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B}) \Big\},$$

since the prior $p(T, B) = 1/|\mathcal{G}|$ and the normalising constant of the Gaussian do not depend on (T, B).

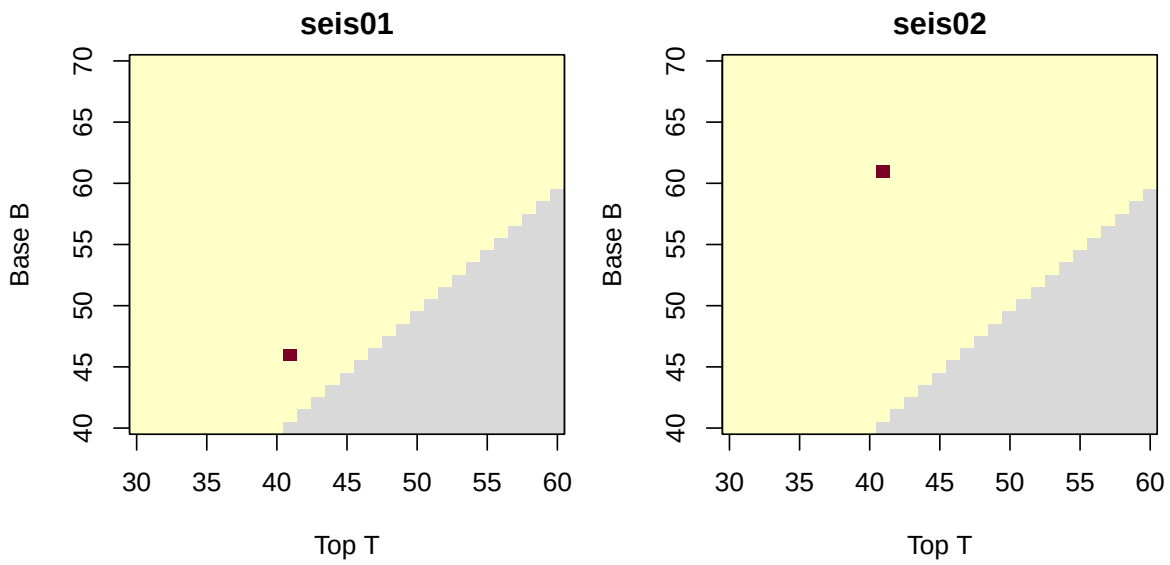


Figure 6: Joint posterior $p(T, B \mid \mathbf{d})$ for seis01 (left) and seis02 (right). Grey: inadmissible pairs.

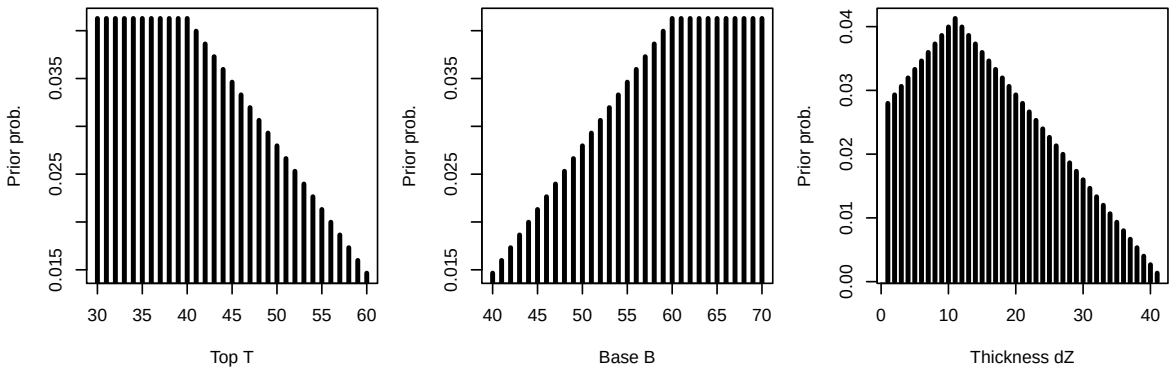


Figure 7: Marginal prior of top T, base B and thickness ΔZ (uniform prior on $\mathcal{G}$).

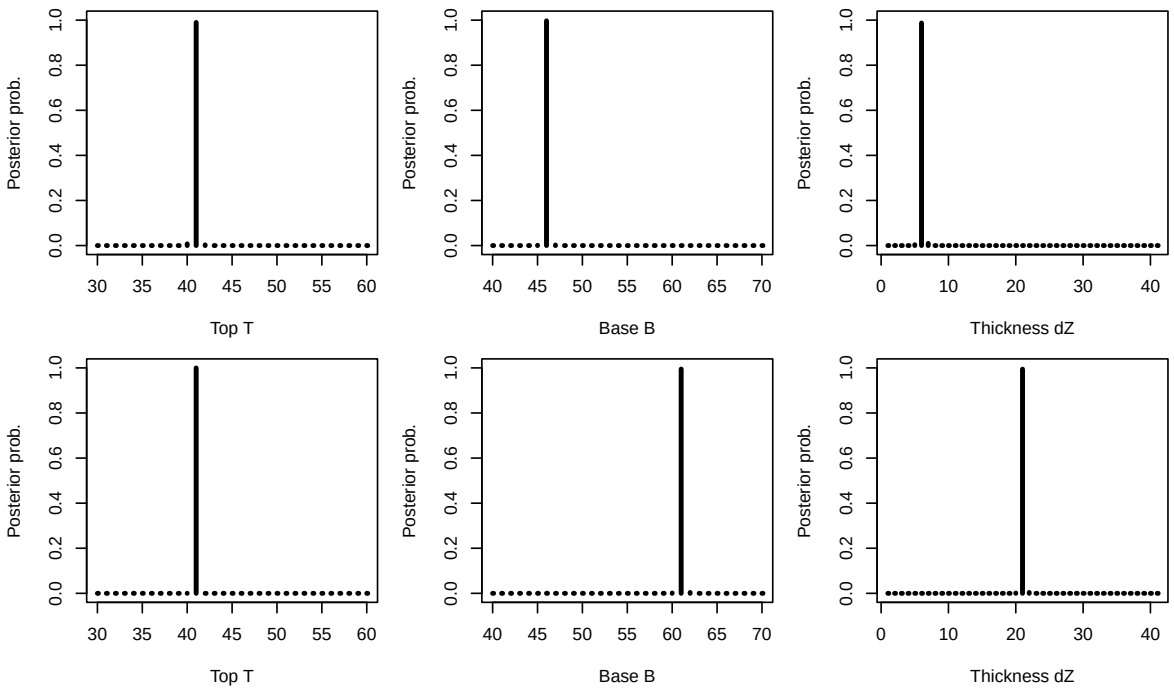


Figure 8: Marginal posterior of T, B and ΔZ. Top row: seis01, bottom row: seis02.

The prior marginals are not uniform, because the number of admissible partners varies with T, B and ΔZ. The posterior is concentrated on $(T, B) = (41, 46)$, $\Delta Z = 6$ for seis01 (probability 0.987) and $(41, 61)$, $\Delta Z = 21$ for seis02 (probability 0.995). The data determine the zone almost exactly, and the prior has little influence.

2b)

The no-zone case is one model, while the zone case consists of $|\mathcal{G}| = 751$ models. Equal prior probability for all models would give $P(\text{no zone}) = 1/752$, i.e. near certainty of a zone only because there are many ways to place it. The prior for the zero case should reflect our belief in whether a zone exists at all. We use

$$P(\text{no zone}) = \tfrac{1}{2}, \qquad p(T, B) = \tfrac{1}{2} \cdot \tfrac{1}{|\mathcal{G}|},\ (T, B) \in \mathcal{G},$$

$$P(\text{no zone} \mid \mathbf{d}) = \frac{\frac{1}{2}\, p(\mathbf{d} \mid \text{no zone})}{\frac{1}{2}\, p(\mathbf{d} \mid \text{no zone}) + \frac{1}{2}\, \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} p(\mathbf{d} \mid T, B)}.$$

$P(\text{no zone} \mid \mathbf{d}) = 2.1 \times 10^{-8}$ for seis01 and 3.4×10^{-13} for seis02.

2c)

Let the action be a_1 (invest) or a_0 (do not invest), with loss equal to minus the profit:

$$L(a_1, V) = -(aV - b), \qquad L(a_0, V) = 0.$$

The Bayes decision minimises the posterior expected loss: $E\{L(a_1, V) \mid \mathbf{d}\} = -(aE(V \mid \mathbf{d}) - b)$ and $E\{L(a_0, V) \mid \mathbf{d}\} = 0$. Hence

$$\text{invest} \iff E(V \mid \mathbf{d}) > \frac{b}{a} = 10.$$

$E(V \mid \mathbf{d})$ is computed by simulation: draw the model (no zone or (T, B)) from its posterior (2b), then $\mathbf{m} \mid \mathbf{d}, T, B$ from 1a with prior mean $\boldsymbol{\mu}_m^{T,B}$, and compute $V = h(\mathbf{m})$.

Table 2: Posterior summaries of V and the profit $aV - b$ (5000 simulations).

	E(V|d)	E(profit|d)	P(profit<0|d)	P10	P50	P90
seis01	31.17	21.17	0	11.66	19.80	32.75
seis02	89.56	79.56	0	60.69	78.71	99.45

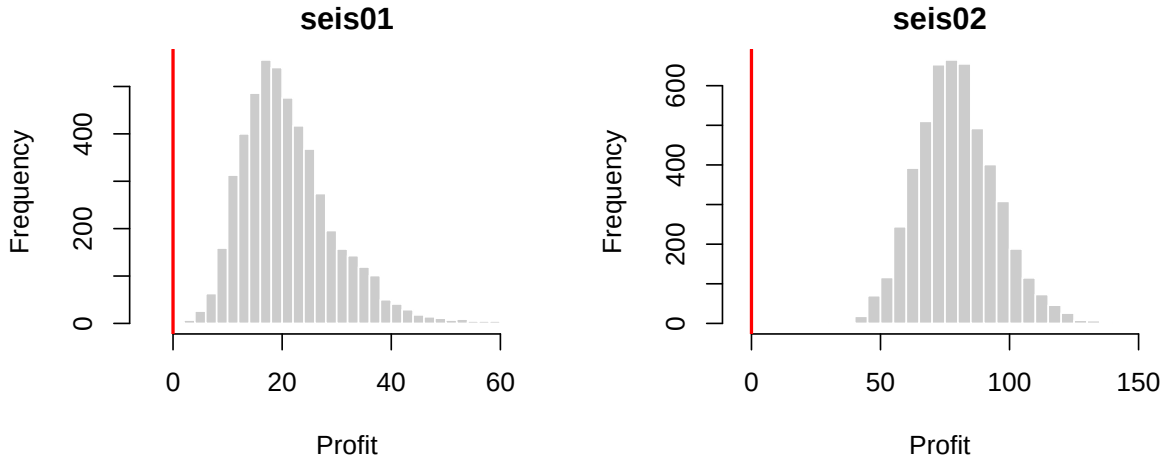


Figure 9: Posterior distribution of the profit $V - 10$. Red line: break-even.

Recommendation: invest in both cases, since $E(V \mid \mathbf{d})$ is far above $b/a = 10$. None of the 5000 simulations gives a loss. The uncertainty is presented by the profit distribution (histograms), its P_{10}, P_{50}, P_{90} and the probability of a loss.

Part 3

3a)

Write $\mathbf{m} = \mu\mathbf{1} - \Delta\mu\,\mathbf{1}_{[T,B]} + \boldsymbol{x}$ with $\boldsymbol{x} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_m)$ independent of $\mu \sim N(\mu_0, \tau_\mu^2)$. By double expectation and total variance,

$$E(\mathbf{m}) = E\{E(\mathbf{m} \mid \mu)\} = \mu_0\mathbf{1} - \Delta\mu\,\mathbf{1}_{[T,B]},$$
$$\mathrm{Cov}(\mathbf{m}) = E\{\mathrm{Cov}(\mathbf{m} \mid \mu)\} + \mathrm{Cov}\{E(\mathbf{m} \mid \mu)\} = \boldsymbol{\Sigma}_m + \tau_\mu^2\mathbf{1}\mathbf{1}^\top,$$

and therefore

$$E(\mathbf{d}) = \mathbf{A}E(\mathbf{m}), \qquad \mathrm{Cov}(\mathbf{d}) = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \tau_\mu^2(\mathbf{A}\mathbf{1})(\mathbf{A}\mathbf{1})^\top + \sigma_\varepsilon^2\mathbf{I}.$$

With $\mu_0 = 0$ the mean is unchanged. Since the rows of $\mathbf{A}$ sum to zero, $\mathbf{A}\mathbf{1} = \mathbf{0}$, so the extra term vanishes and $\mathrm{Cov}(\mathbf{d})$ is also unchanged. Hence $p(\mathbf{d} \mid T, B)$, and so $p(T, B \mid \mathbf{d})$, are exactly the same as in 2a.

Recomputing $p(T, B \mid \mathbf{d})$ for seis01 gives the same posterior as in 2a (largest difference 2.6×10^{-16}). This is not surprising: the data contain no information about a common level of $\mathbf{m}$.

3b)

With $\sigma_\varepsilon^2 = \nu\sigma_m^2$ and $\mathbf{R} = \boldsymbol{\Sigma}_m/\sigma_m^2$ (the correlation matrix), $\mathrm{Cov}(\mathbf{d}) = \sigma_m^2\mathbf{C}$ with $\mathbf{C} = \mathbf{A}\mathbf{R}\mathbf{A}^\top + \nu\mathbf{I}$, so for a given (T, B)

$$p(\mathbf{d} \mid \sigma_m^2, T, B) \propto (\sigma_m^2)^{-n/2} \exp\Big\{-\frac{Q_{T,B}}{2\sigma_m^2}\Big\}, \qquad Q_{T,B} = (\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B})^\top\mathbf{C}^{-1}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B}).$$

The conjugate prior is $\sigma_m^2 \sim \text{IG}(\alpha, \beta)$, and then

$$p(\sigma_m^2 \mid \mathbf{d}, T, B) \propto (\sigma_m^2)^{-(\alpha+n/2)-1} \exp\Big\{ -\frac{\beta + Q_{T,B}/2}{\sigma_m^2} \Big\} \;\Rightarrow\; \sigma_m^2 \mid \mathbf{d}, T, B \sim \text{IG}\big(\alpha + \tfrac{n}{2},\; \beta + \tfrac{1}{2} Q_{T,B}\big).$$

Since (T, B) is unknown, $p(\sigma_m^2 \mid \mathbf{d}) = \sum_{\mathcal{G}} p(T, B \mid \mathbf{d})\, \text{IG}(\sigma_m^2; \alpha + \frac{n}{2}, \beta + \frac{1}{2} Q_{T,B})$, with $p(T, B \mid \mathbf{d})$ from 3c. We use $\alpha = 2$, $\beta = 1$ (prior mean 1) and $\nu = 0.04$.

Table 3: Posterior mean and 95% credibility interval for σ_m^2.

	posterior mean	2.5%	97.5%
seis01	1.27	0.97	1.66
seis02	1.05	0.79	1.37

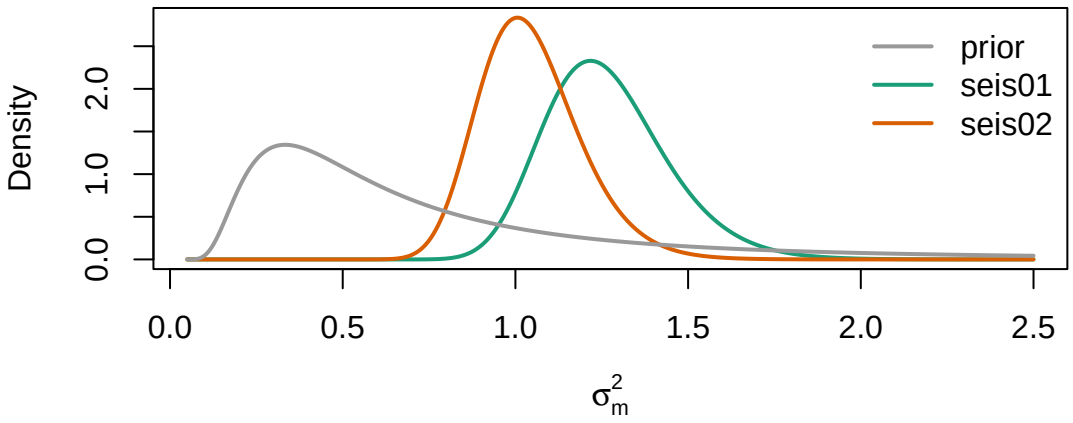


Figure 10: Prior IG(2,1) (grey) and posterior densities of σ_m^2 for seis01 and seis02.

3c)

The uncertainty is accounted for by integrating σ_m^2 out:

$$p(\mathbf{d} \mid T, B) = \int p(\mathbf{d} \mid \sigma_m^2, T, B)\, p(\sigma_m^2)\, d\sigma_m^2 \;\propto\; \big(\beta + \tfrac{1}{2} Q_{T,B}\big)^{-(\alpha+n/2)}$$

(a multivariate Student-t density), so $p(T, B \mid \mathbf{d}) \propto (\beta + \frac{1}{2} Q_{T,B})^{-(\alpha+n/2)}$, compared with $\propto \exp(-\frac{1}{2} Q_{T,B})$ for fixed $\sigma_m^2 = 1$.

Table 4: Posterior of (T, B) with fixed $\sigma_m^2 = 1$ (columns 1–3) and with σ_m^2 integrated out (columns 4–6).

	mode	prob.	E(dZ|d)	mode	prob.	E(dZ|d)
seis01	(41,46)	0.987	6.01	(41,46)	0.954	6.02

	mode	prob.	E(dZ|d)	mode	prob.	E(dZ|d)
seis02	(41,61)	0.995	21	(41,61)	0.991	21

The most probable pair is unchanged, but the posterior is slightly wider: the probability of the mode falls from 0.987 to 0.954 (seis01) and from 0.995 to 0.991 (seis02).

Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
A  <- as.matrix(read.table("data/A.txt"))   # forward operator (n x n)
dimnames(A) <- NULL
d0 <- scan("data/seis00.txt", quiet = TRUE)
d1 <- scan("data/seis01.txt", quiet = TRUE)
d2 <- scan("data/seis02.txt", quiet = TRUE)
n  <- nrow(A)
z  <- 1:n
# Prior covariance: exponential correlation function
exp_cov <- function(n, sigma2, ell) {
  D <- abs(outer(1:n, 1:n, "-"))           # matrix of |i - j|
  sigma2 * exp(-3 * D / ell)
}

# Posterior of m | d for d = A m + eps (Gaussian conditioning)
gauss_posterior <- function(d, A, mu, Sigma, sigma2_eps) {
  Sigma_d <- A %*% Sigma %*% t(A) + sigma2_eps * diag(nrow(A))   # Cov(d)
  K       <- Sigma %*% t(A) %*% solve(Sigma_d)                   # Cov(m,d) Cov(d)^-1
  mu_post <- drop(mu + K %*% (d - A %*% mu))
  S_post  <- Sigma - K %*% A %*% Sigma
  S_post  <- (S_post + t(S_post)) / 2      # make exactly symmetric (rounding)
  list(mean = mu_post, cov = S_post)
}

# Simulate B draws from N(mu, S): m = mu + L x, L L^T = S, x ~ N(0, I)
rmvn <- function(B, mu, S) {
  L <- t(chol(S + 1e-10 * diag(length(mu))))   # lower Cholesky factor (tiny jitter)
  X <- matrix(rnorm(length(mu) * B), length(mu), B)
  t(mu + L %*% X)                               # one draw per row
}

# Mean with pointwise 95% credibility band (and optional realisations)
band_plot <- function(mean, sd, main, ylim, sims = NULL) {
  lo <- mean - 1.96 * sd
  hi <- mean + 1.96 * sd
  plot(z, mean, type = "n", ylim = ylim, xlab = "Index i", ylab = "m_i", main = main)
  polygon(c(z, rev(z)), c(lo, rev(hi)), col = "grey85", border = NA)
  if (!is.null(sims)) matlines(z, t(sims), lty = 1, lwd = 1,
                               col = c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e"))
  lines(z, mean, lwd = 2)
}
mu <- 0; sigma2_m <- 1; ell <- 20; sigma2_eps <- 0.04
mu_m    <- rep(mu, n)
Sigma_m <- exp_cov(n, sigma2_m, ell)

set.seed(1)
prior_sims <- rmvn(5, mu_m, Sigma_m)
band_plot(mu_m, sqrt(diag(Sigma_m)), "Prior", ylim = c(-3.5, 3.5), sims = prior_sims)
post0 <- gauss_posterior(d0, A, mu_m, Sigma_m, sigma2_eps)
```

```r
sd0   <- sqrt(diag(post0$cov))
band_plot(post0$mean, sd0, "Posterior given seis00", ylim = c(-3.5, 3.5))
abline(h = c(-1.96, 1.96), lty = 2, col = "grey40")
cov_image <- function(S, main, zlim) {
  image(z, z, t(S)[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "j", ylab = "i", main = main, axes = FALSE, useRaster = TRUE)
  axis(1); axis(2, at = n + 1 - c(1, 20, 40, 60, 80, 100), labels = c(1, 20, 40, 60, 80, 100))
  box()
}
zl <- range(Sigma_m, post0$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cov_image(Sigma_m, "Prior covariance", zl)
cov_image(post0$cov, "Posterior covariance", zl)
c_lev <- -1
h_V   <- function(m, c = c_lev) sum((c - m) * (m < c))   # exceedance volume

B <- 100
set.seed(2)
post_sims <- rmvn(B, post0$mean, post0$cov)
V <- apply(post_sims, 1, h_V)                              # one V per posterior draw

V_summary <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_summary, 2)), caption = "Posterior summaries of $V$ ($B=100$).")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
Vs  <- sort(V)                                          # order statistics
P10 <- Vs[10]
P90 <- Vs[90]

# N = number of samples below the true P10 ~ Bin(B, 0.1)
# V_(r) <= P10 < V_(s)  <=>  r <= N <= s - 1
r <- qbinom(0.025, B, 0.1)
s <- qbinom(0.975, B, 0.1) + 1
coverage <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)
lag1 <- function(x) cor(x[-1], x[-length(x)])   # lag-1 correlation

rho_obs <- lag1(d0)
set.seed(4)
rho_perm <- replicate(1000, lag1(sample(d0)))    # sample() gives a random permutation
n_extreme <- sum(abs(rho_perm) >= abs(rho_obs))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = c(-0.4, 1))
abline(v = rho_obs, col = "red", lwd = 2)
dmu <- 5
G   <- subset(expand.grid(T = 30:60, B = 40:70), T <= B)    # admissible (T, B)
G$dZ <- G$B - G$T + 1                                       # thickness

# Prior mean for a zone from T to B (T = NA: no zone)
mu_TB <- function(T, B) {
  m <- rep(mu, n)
  if (!is.na(T)) m[T:B] <- mu - dmu
  m
}
```

```r
Sigma_d <- A %*% Sigma_m %*% t(A) + sigma2_eps * diag(n)
L_d     <- t(chol(Sigma_d))                                  # Sigma_d = L_d L_d^T

# log p(d | mean) up to a constant common to all models
loglik <- function(d, mean_m) {
  r <- forwardsolve(L_d, d - A %*% mean_m)                   # L_d^-1 (d - A mu)
  -0.5 * sum(r^2)
}

log_sum_exp <- function(x) max(x) + log(sum(exp(x - max(x))))

zone_posterior <- function(d) {
  ll <- sapply(seq_len(nrow(G)), function(k) loglik(d, mu_TB(G$T[k], G$B[k])))
  list(ll = ll, post = exp(ll - log_sum_exp(ll)))            # uniform prior cancels
}

zp1 <- zone_posterior(d1)
zp2 <- zone_posterior(d2)
heat <- function(post, main) {
  M <- matrix(NA, 31, 31, dimnames = list(30:60, 40:70))
  M[cbind(G$T - 29, G$B - 39)] <- post
  image(30:60, 40:70, M, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "Top T", ylab = "Base B", main = main)
  image(30:60, 40:70, ifelse(is.na(M), 1, NA), col = "grey85", add = TRUE)
  box()
}
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
heat(zp1$post, "seis01")
heat(zp2$post, "seis02")
marg <- function(w, v, levels) sapply(levels, function(x) sum(w[v == x]))
prior_G <- rep(1 / nrow(G), nrow(G))
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(30:60, marg(prior_G, G$T,  30:60), type = "h", lwd = 3, xlab = "Top T",  ylab = "Prior prob.")
plot(40:70, marg(prior_G, G$B,  40:70), type = "h", lwd = 3, xlab = "Base B", ylab = "Prior prob.")
plot(1:41,  marg(prior_G, G$dZ, 1:41),  type = "h", lwd = 3, xlab = "Thickness dZ", ylab = "Prior prob.
par(mfrow = c(2, 3), mar = c(4, 4, 1, 1))
for (zp in list(zp1, zp2)) {
  plot(30:60, marg(zp$post, G$T,  30:60), type = "h", lwd = 3, ylim = c(0, 1), xlab = "Top T",  ylab =
  plot(40:70, marg(zp$post, G$B,  40:70), type = "h", lwd = 3, ylim = c(0, 1), xlab = "Base B", ylab =
  plot(1:41,  marg(zp$post, G$dZ, 1:41),  type = "h", lwd = 3, ylim = c(0, 1), xlab = "Thickness dZ", yl
}
summ <- function(post) {
  k <- which.max(post)
  c(T = G$T[k], B = G$B[k], dZ = G$dZ[k], prob = post[k], E_dZ = sum(post * G$dZ))
}
tab2a <- rbind(seis01 = summ(zp1$post), seis02 = summ(zp2$post))
p0 <- 1 / 2
post_zero <- function(zp, d) {
  l0 <- log(p0)     + loglik(d, mu_TB(NA, NA))
  lz <- log(1 - p0) + log_sum_exp(zp$ll) - log(nrow(G))
  exp(l0 - log_sum_exp(c(l0, lz)))
}
pz <- c(seis01 = post_zero(zp1, d1), seis02 = post_zero(zp2, d2))
```

```r
a <- 1; b <- 10; B_mc <- 5000
K_gain <- Sigma_m %*% t(A) %*% solve(Sigma_d)
L_post <- t(chol(post0$cov + 1e-10 * diag(n)))            # same posterior covariance as 1c

sim_V <- function(zp, d, pz0) {
  w   <- c(pz0, (1 - pz0) * zp$post)                      # no zone + all (T, B)
  Tm  <- c(NA, G$T); Bm <- c(NA, G$B)
  idx <- sample(length(w), B_mc, replace = TRUE, prob = w)
  sapply(idx, function(k) {
    mu_k <- mu_TB(Tm[k], Bm[k])
    m    <- mu_k + K_gain %*% (d - A %*% mu_k) + L_post %*% rnorm(n)
    h_V(m)
  })
}
set.seed(5)
V1 <- sim_V(zp1, d1, pz[1])
V2 <- sim_V(zp2, d2, pz[2])
profit <- list(seis01 = a * V1 - b, seis02 = a * V2 - b)

tab2c <- t(sapply(profit, function(x)
  c(EV = mean(x + b) / a, E_profit = mean(x), P_loss = mean(x < 0),
    P10 = unname(quantile(x, 0.1)), P50 = median(x), P90 = unname(quantile(x, 0.9)))))
knitr::kable(tab2c, digits = 2,
             col.names = c("E(V|d)", "E(profit|d)", "P(profit<0|d)", "P10", "P50", "P90"),
             caption = "Posterior summaries of $V$ and the profit $aV-b$ (5000 simulations).")
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for (nm in names(profit)) {
  hist(profit[[nm]], breaks = 30, col = "grey80", border = "white", main = nm,
       xlab = "Profit", xlim = range(-5, profit[[nm]]))
  abline(v = 0, col = "red", lwd = 2)
}
tau2_mu <- 0.25
Sigma_m_mu <- Sigma_m + tau2_mu * matrix(1, n, n)            # Cov(m) with uncertain mu
Sigma_d_mu <- A %*% Sigma_m_mu %*% t(A) + sigma2_eps * diag(n)
L_d_save <- L_d
L_d <- t(chol(Sigma_d_mu))                                    # loglik() now uses new Cov(d)
zp1_mu <- zone_posterior(d1)
L_d <- L_d_save                                               # restore
diff3a <- c(cov_d = max(abs(Sigma_d_mu - Sigma_d)), post = max(abs(zp1_mu$post - zp1$post)))
alpha <- 2; beta <- 1; nu <- 0.04
C_mat <- A %*% (Sigma_m / sigma2_m) %*% t(A) + nu * diag(n)
L_C   <- t(chol(C_mat))
Q_TB  <- function(d) sapply(seq_len(nrow(G)), function(k) {
  r <- forwardsolve(L_C, d - A %*% mu_TB(G$T[k], G$B[k]))
  sum(r^2)
})

# 3c: p(T,B | d) with sigma_m^2 integrated out (Student-t marginal likelihood)
zone_posterior_t <- function(Q) {
  lw <- -(alpha + n / 2) * log(beta + Q / 2)
  exp(lw - log_sum_exp(lw))
}
Q1 <- Q_TB(d1); Q2 <- Q_TB(d2)
```

```r
pt1 <- zone_posterior_t(Q1); pt2 <- zone_posterior_t(Q2)

# Posterior of sigma_m^2: mixture of inverse-gammas, summarised by simulation
sim_sigma2 <- function(Q, w, N = 20000) {
  k <- sample(length(w), N, replace = TRUE, prob = w)
  1 / rgamma(N, shape = alpha + n / 2, rate = beta + Q[k] / 2)
}
set.seed(6)
s2_1 <- sim_sigma2(Q1, pt1); s2_2 <- sim_sigma2(Q2, pt2)
tab3b <- rbind(seis01 = c(mean(s2_1), quantile(s2_1, c(0.025, 0.975))),
               seis02 = c(mean(s2_2), quantile(s2_2, c(0.025, 0.975))))
knitr::kable(tab3b, digits = 2, col.names = c("posterior mean", "2.5%", "97.5%"),
             caption = "Posterior mean and 95\\% credibility interval for $\\sigma_m^2$.")
dinvgamma <- function(x, a, b) exp(a * log(b) - lgamma(a) - (a + 1) * log(x) - b / x)
mix_dens  <- function(x, Q, w) sapply(x, function(s)
  sum(w * dinvgamma(s, alpha + n / 2, beta + Q / 2)))
xs <- seq(0.05, 2.5, length.out = 400)
plot(xs, mix_dens(xs, Q1, pt1), type = "l", lwd = 2, col = "#1b9e77",
     xlab = expression(sigma[m]^2), ylab = "Density",
     ylim = c(0, max(mix_dens(xs, Q2, pt2), mix_dens(xs, Q1, pt1))))
lines(xs, mix_dens(xs, Q2, pt2), lwd = 2, col = "#d95f02")
lines(xs, dinvgamma(xs, alpha, beta), lwd = 2, col = "grey60")
legend("topright", c("prior", "seis01", "seis02"), lwd = 2,
       col = c("grey60", "#1b9e77", "#d95f02"), bty = "n")
cmp <- function(pf, pt) {
  kf <- which.max(pf); kt <- which.max(pt)
  c(mode_fixed = paste0("(", G$T[kf], ",", G$B[kf], ")"), prob_fixed = round(pf[kf], 3),
    E_dZ_fixed = round(sum(pf * G$dZ), 2),
    mode_t = paste0("(", G$T[kt], ",", G$B[kt], ")"), prob_t = round(pt[kt], 3),
    E_dZ_t = round(sum(pt * G$dZ), 2))
}
tab3c <- rbind(seis01 = cmp(zp1$post, pt1), seis02 = cmp(zp2$post, pt2))
knitr::kable(tab3c, col.names = c("mode", "prob.", "E(dZ|d)", "mode", "prob.", "E(dZ|d)"),
             caption = "Posterior of $(T,B)$ with fixed $\\sigma_m^2=1$ (columns 1--3) and with $\\sigma
```